

HDOC DAY-21 ACTIVITY

QUESTION 1 – Menu-Driven Tables for Integers

Write a menu-driven C program that accepts two positive integers, m and n , where $m \leq n$. Display the following menu and accept the user's choice. Use a switch statement to perform the selected operation:

1. Print the first 10 multiples of each integer from m to n .
2. Print the addition table of each integer from m to n , using addends from 1 to 10.
3. Print the subtraction table of each integer from m to n , using subtrahends from 1 to 10.
4. Print the results obtained by dividing each integer from m to n by the integers from 1 to 10. Display the results rounded to two decimal places.
5. Exit the program.

Display an appropriate error message if the user enters an invalid menu choice. Continue displaying the menu and performing the selected operation until the user chooses the Exit option.

Note: Prepare the algorithm and test-case table. Then write, compile and execute the C program, and test it using the prepared test cases.

1. Problem Understanding

- Input: Two positive integers m and n , where $m \leq n$, and the user's menu choice.
- Output: The result of the operation selected through switch-case.
- The menu is displayed repeatedly until the user selects Exit.
- Division results are displayed rounded to two decimal places.

Operation Summary

Choice	Operation	Main Logic
1	First 10 multiples	Use a loop from 1 to 10 and calculate number $\times$ multiplier.
2	Addition table	Use addends from 1 to 10 and calculate number $+$ addend.
3	Subtraction table	Use subtrahends from 1 to 10 and calculate number $-$ subtrahend.
4	Division results	Use divisors from 1 to 10 and calculate number $/$ divisor.
5	Exit	Terminate the program.
Other	Invalid choice	Display an appropriate error message.

2. Standard Test Cases

Test Case	Choice	Input	Expected Output	Actual Output	Result
1	1	$m=2, n=3$	Multiples of 2 and 3	Multiples of 2 and 3	Pass
2	2	$m=2, n=3$	Addition tables for 2 and 3	Addition tables for 2 and 3	Pass

Test Case	Choice	Input	Expected Output	Actual Output	Result
3	3	m=2, n=3	Subtraction tables for 2 and 3	Subtraction tables for 2 and 3	Pass
4	4	m=2, n=3	Division results for 2 and 3	Division results for 2 and 3	Pass
5	5	m=2, n=3	Program exits	Program exits	Pass
6	1	m=5, n=5	Multiples of 5	Multiples of 5	Pass
7	4	m=1, n=1	Division results for 1	Division results for 1	Pass
8	2	m=5, n=3	Error: m must be $\leq$ n	Error: m must be $\leq$ n	Pass
9	6	m=2, n=3	Invalid choice	Invalid choice	Pass

3. Algorithm

1. Start the program.
2. Declare the required variables m, n, choice, i, and num.
3. Read m and n.
4. Validate m and n. If either is not positive, display an appropriate error message and request the values again.
5. Validate that $m \leq n$. If m is greater than n, display an appropriate error message and request the values again.
6. Repeat the following steps until the user chooses Exit.
7. Display the menu and read the user's choice.
8. Use switch-case to perform the selected operation.
9. Case 1: For each number from m to n, use a loop from 1 to 10 and display number $\times$ i.
10. Case 2: For each number from m to n, use a loop from 1 to 10 and display number $+$ i.
11. Case 3: For each number from m to n, use a loop from 1 to 10 and display number $-$ i.
12. Case 4: For each number from m to n, use a loop from 1 to 10 and display number $/$ i rounded to two decimal places.
13. Case 5: Display an exit message and terminate the menu loop.
14. Default: Display an invalid choice message.
15. After each operation except Exit, display the menu again.
16. Stop the program.

4. C Program

```
#include <stdio.h>

int main()
{
    int m, n, choice, i, num;
    do
    {
        printf("Enter m: ");
```

```

scanf("%d", &m);
printf("Enter n: ");
scanf("%d", &n);
if (m <= 0 || n <= 0 || m > n)
    printf("Invalid input. Enter positive values with m <= n.\n");
}
while (m <= 0 || n <= 0 || m > n);
do
{
    printf("\nMenu\n");
    printf("1. Print first 10 multiples\n");
    printf("2. Print addition table\n");
    printf("3. Print subtraction table\n");
    printf("4. Print division results\n");
    printf("5. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            for (num = m; num <= n; num++)
            {
                printf("\nMultiples of %d:\n", num);
                for (i = 1; i <= 10; i++)
                    printf("%d x %d = %d\n", num, i, num * i);
            }
            break;
        case 2:
            for (num = m; num <= n; num++)
            {
                printf("\nAddition table of %d:\n", num);
                for (i = 1; i <= 10; i++)
                    printf("%d + %d = %d\n", num, i, num + i);
            }
            break;
        case 3:
            for (num = m; num <= n; num++)
            {
                printf("\nSubtraction table of %d:\n", num);

```

```

        for (i = 1; i <= 10; i++)
            printf("%d - %d = %d\n", num, i, num - i);
    }
    break;
case 4:
    for (num = m; num <= n; num++)
    {
        printf("\nDivision results for %d:\n", num);
        for (i = 1; i <= 10; i++)
            printf("%d / %d = %.2f\n", num, i, (float)num / i);
    }
    break;
case 5:
    printf("Exiting the program.\n");
    break;
default:
    printf("Invalid choice.\n");
}
}
while (choice != 5);
return 0;
}

```

5. Compilation and Execution Evidence



```

kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)~[~/Desktop]
$ gedit question.c
** (gedit:2464): WARNING **: 05:18:59.174: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2464): WARNING **: 05:18:59.174: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
(kali@kali)~[~/Desktop]
$ gcc question.c -o question
(kali@kali)~[~/Desktop]
$ ./question
Enter m: 2
Enter n: 3
Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 1
Multiples of 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20
Multiples of 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12

```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 2

Addition table of 2:
2 + 1 = 3
2 + 2 = 4
2 + 3 = 5
2 + 4 = 6
2 + 5 = 7
2 + 6 = 8
2 + 7 = 9
2 + 8 = 10
2 + 9 = 11
2 + 10 = 12

Addition table of 3:
3 + 1 = 4
3 + 2 = 5
3 + 3 = 6
3 + 4 = 7
3 + 5 = 8
3 + 6 = 9
3 + 7 = 10
3 + 8 = 11
3 + 9 = 12
3 + 10 = 13
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
3 + 10 = 13

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 3

Subtraction table of 2:
2 - 1 = 1
2 - 2 = 0
2 - 3 = -1
2 - 4 = -2
2 - 5 = -3
2 - 6 = -4
2 - 7 = -5
2 - 8 = -6
2 - 9 = -7
2 - 10 = -8

Subtraction table of 3:
3 - 1 = 2
3 - 2 = 1
3 - 3 = 0
3 - 4 = -1
3 - 5 = -2
3 - 6 = -3
3 - 7 = -4
3 - 8 = -5
3 - 9 = -6
3 - 10 = -7

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 4
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
5. Exit
Enter your choice: 4

Division results for 2:
2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
2 / 5 = 0.40
2 / 6 = 0.33
2 / 7 = 0.29
2 / 8 = 0.25
2 / 9 = 0.22
2 / 10 = 0.20

Division results for 3:
3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
3 / 4 = 0.75
3 / 5 = 0.60
3 / 6 = 0.50
3 / 7 = 0.43
3 / 8 = 0.38
3 / 9 = 0.33
3 / 10 = 0.30

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 5
Exiting the program.

(kali@kali)~[/Desktop]
$ ./question
Enter m: 5
Enter n: 5
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
Enter n: 5

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 1

Multiples of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 5
Exiting the program.

(kali@kali)~[/Desktop]
$ ./question
Enter m: 1
Enter n: 1

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 4

Division results for 1:
1 / 1 = 1.00
1 / 2 = 0.50
1 / 3 = 0.33
1 / 4 = 0.25
1 / 5 = 0.20
1 / 6 = 0.17
1 / 7 = 0.14
1 / 8 = 0.12
1 / 9 = 0.11
1 / 10 = 0.10

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 5
Exiting the program.

(kali@kali)~/Desktop
$ ./question
Enter m: 5
Enter n: 3
Invalid input. Enter positive values with m ≤ n.
Enter m: 5
Enter n: 5

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 5
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
Enter m: 5
Enter n: 3
Invalid input. Enter positive values with m ≤ n.
Enter m: 5
Enter n: 5

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 5
Exiting the program.

(kali@kali)~/Desktop
$ ./question
Enter m: 2
Enter n: 3

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 6
Invalid choice.

Menu
1. Print first 10 multiples
2. Print addition table
3. Print subtraction table
4. Print division results
5. Exit
Enter your choice: 5
Exiting the program.

(kali@kali)~/Desktop
$
```

6. Verification of All Test Cases

Test Case	Choice	Input	Expected Output	Actual Output	Result
1	1	m=2, n=3	Multiples of 2 and 3	Multiples of 2 and 3	Pass
2	2	m=2, n=3	Addition tables for 2 and 3	Addition tables for 2 and 3	Pass
3	3	m=2, n=3	Subtraction tables for 2 and 3	Subtraction tables for 2 and 3	Pass
4	4	m=2, n=3	Division results for 2 and 3	Division results for 2 and 3	Pass
5	5	m=2, n=3	Program exits	Program exits	Pass
6	1	m=5, n=5	Multiples of 5	Multiples of 5	Pass

Test Case	Choice	Input	Expected Output	Actual Output	Result
7	4	m=1, n=1	Division results for 1	Division results for 1	Pass
8	2	m=5, n=3	Error: m must be $\leq$ n	Error: m must be $\leq$ n	Pass
9	6	m=2, n=3	Invalid choice	Invalid choice	Pass